

Supplementary Material

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1 Dataset

1.1 Chess Dataset

This section describes the datasets used in applications. In the chess application, we obtained the dataset of chess games from Mega ChessBase 2025 [ChessBase GmbH, 2024] by purchasing it. It contains more than 113,000 expert-annotated games and high-class analysed games. The dataset includes the names of two players, their ranks, their moves, the total number of moves, the year the game occurred, the game results, the annotator’s name, and the comments, if provided. The dataset is rich, but not all attributes are relevant to our research.

1.2 Artwork Dataset

In the artwork application, we obtained the fine-art auction data, which was reported to be anonymous as of September 2nd, 2025. The dataset contains essential attributes, such as lot ID, artwork ID, artist ID, sale date, artwork name, artist name, artist nationality, artist birth and death years, artwork creation year, estimated auction price range, and sale price in both local currency and US dollars. Initially, the dataset contains 1,552,843 data points on the auctions of paintings by 1,552 artists, with some paintings auctioned multiple times. As stated in the main paper, we removed dubious records, such as auction data without a valid auction image and artwork data without a creation year, and limited our artists to Western European artists, removing samples with years that appear incorrect. For example, artworks recorded as 18984 year or artworks recorded as 6 A.D.. Also, the artwork’s creation year should fall within the artist’s lifetime. In art history, the Medieval period spans from around 500 to 1400 A.D. Therefore, we retain only artworks dated between 500 A.D. and 2024. The artwork sold in 2025 was removed because the Consumer Price Index (CPI) for 2025 was not released when inflation was computed. Finally, there are 358,867 auction data points. There are 355,551 unique artworks. Among them, 2,802 artworks have multiple auction records, with a maximum of 4. These are the final dataset pools to compute newness, surprise, and value for artworks.

Because of the purchase contract, we are unable to publish the dataset.

2 Labeling

When labeling based on expert comments with LLMs in the chess application, we used synonyms of multiple words related to ‘creative’ to filter for comments related to creativity. According to the Merriam-Webster Dictionary, synonyms of the word ‘creative’ include ‘innovative’, ‘inventive’, ‘imaginative’, and ‘talented’. Then, the most related synonyms of ‘new’ are ‘novel’, ‘unfamiliar’, ‘fresh’, ‘strange’, ‘unprecedented’, and ‘original’. The synonyms of ‘surprising’ are ‘startling’, ‘amazing’, ‘shocking’, and ‘stunning’. Furthermore, when manually reviewing the comments that explicitly mention ‘creative’, we find that ‘brilliant’, ‘inspiration’, ‘unusual’, and ‘impressive’ are also important keywords for us. We identified about 17,000 comments using the keywords above. Then we applied the prompts in the following section to ask for labelling.

3 Prompts

The following are the prompts for comment labelling in the chess application and artwork criticism in the artwork application.

Prompt for Chess Comment Labelling

You are a Chess expert and diligent annotator. Answer three questions based on a given expert comment about a chess position/move sequence. Use ONLY the supplied inputs (move number, FEN, comment). Do not infer from outside knowledge.

TASKS

1) What is the comment on? Choose ONE:

- “move”: comment targets a single move (e.g., “e2-e4”, “...Nd5?”).
- “sequence”: comment targets a sequence leading to the current position (plans, manoeuvres, forcing line).
- “opening”: comment is about the first 12–18 moves or opening theory (“In the Sicilian...”, “This line...”).
- “game”: comment evaluates the whole game (“In this game...”, “throughout the game...”).
- “remove”: comment is unrelated to the current

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game (history, player biography, unrelated historical games).

- “not sure”: cannot decide; needs human review.

2) Is creativity (novelty + value + surprise) being asserted or implied?

Choose ONE:

- “positive”: says/implies the move(s) are creative (“inventive”, “impressive”).

- “negative”: says/implies NOT creative/ordinary/standard (“typical”, “theory”, “routine”, “book move”).

- “not related”: creativity isn’t discussed.

- “refer”: credits creativity to others (“novel idea by X”, “the innovation originated with Y”).

- “not sure”: ambiguous; needs review.

3) Are the move(s) described as surprising?

Choose ONE:

- “positive”: surprising in a good sense (“unexpected”, “shock”, “stunning”, “!!” with surprise wording).

- “negative”: explicitly not surprising / very common (“frequently played”, “normal”, “theoretical”).

- “not related”: surprise isn’t discussed.

- “not sure”: ambiguous; needs review.

DECISION RULES

- Prefer “move” when the wording pinpoints a single ply (“Here 17...Nd5!”). Prefer “sequence” if discussing a line, plan, or system.

- Prefer “opening” when referring to early theory or strategy, named openings/variations, or “book” lines.

- Prefer “game” if judging overall game quality/arc/end result without focusing on a move or short line.

- Use “remove” for historical anecdotes/biographical notes not tied to this game/position.

- If unsure, pick “not sure” (do not guess).

- For Q2 (creativity): require explicit or strong implication of novelty, surprise, and goodness. “Theoretical” or “known” → likely “negative”.

- For Q3 (surprise): require explicit surprise terms or clear implication. Absence of such leads to “not related”.

OUTPUT FORMAT

analysis: your explanation here

q1: move

q2: positive

q3: positive

CONSTRAINTS

- Base decisions ONLY on the provided comment text; move number and FEN are for context cues, not engine eval.

- Never reference external databases or the game’s actual theory.

- If the comment mixes topics, choose the primary focus; mention the secondary in analysis.

- When in doubt, prefer “not sure” over speculation.

Prompt for Artwork Criticism

You are an expert in artwork criticism, art history, and creativity evaluation.

Using web search, find authoritative commentary on the artistic value and creative significance of the artwork:

- Title: {content[“title”]}

- Artist: {content[‘first’]} {content[‘last’]}

{content[‘nationality’]}

- Year of Creation: {content[‘workyear from’]}

Based strictly on information gathered through web search, summarize the consensus and key scholarly interpretations in English regarding the artwork’s artistic value and creativity.

Your analysis should address innovations in composition, technique, emotional expression, iconography, and influences on later art.

- An artwork is creative only if it is new, valuable, and surprising compared to prior artworks

OUTPUT FORMAT (around 150 words each):

Artistic Value: <High or Low>

<Your summary here >

Creativity: <Yes or No >

<Your summary here >

References

[ChessBase GmbH, 2024] ChessBase GmbH. Mega chess-base 2025. <https://www.chessbase.com>, 2024. Commercial chess games database.